

Overview

Master Mini is a humanoid robot development platform for competitions, education, entertainment, and related scenarios. It is affordable, lightweight, and durable.

1.1 Product Composition

The MASTER MINI robot consists of the head, torso, arms, and legs. It has 22 degrees of freedom across the whole body, enabling flexible motion and posture control.

- The head has 2 degrees of freedom, including the head yaw joint and head pitch joint. The depth camera and microphone array are located in the head.
- Each arm has 4 degrees of freedom, including the shoulder pitch joint, shoulder roll joint, shoulder yaw joint, and elbow joint.
- Each leg has 6 degrees of freedom, including the hip pitch joint, hip roll joint, hip yaw joint, knee joint, upper ankle joint, and lower ankle joint.
- The torso contains the controller, speaker, and battery.

See details in [Specifications](#)

1.2 What Can It Do?

Motion Capabilities. MASTER MINI supports omnidirectional walking and compound gaits, including forward/backward, lateral, and rotational movement. It can walk on uneven ground, has strong disturbance resistance, and actively recovers balance when subjected to a certain external force.

Built-in Agents. MASTER MINI includes 6 official Agents, which work like ready-to-use "apps" covering entertainment, conversation, teaching, competition, and other scenarios. Developers can also create their own Agents and share them with other users.

Agent	Scenario	Core Capability
Master Mini	Getting Started	A general-purpose basic Agent that supports daily actions such as waving, handshaking, and bowing with folded hands. It can also perform several fun dances for everyday interaction scenarios.
Soccer	Football Competition	Designed for soccer scenarios. It can autonomously track, chase, kick, and shoot the ball, and can accurately reproduce signature celebration moves from soccer stars such as Cristiano Ronaldo and Messi.
Hi Chat	Voice Interaction	An intelligent conversational Agent that can switch flexibly among fun personas such as a cute child, tutor, or pet, meeting needs for casual chat and entertaining companionship. See details in Voice Interaction
Teach	Teaching	A hands-on learning Agent. Users physically guide the robot through a motion, and the device quickly recognizes and accurately reproduces it, making custom actions easy to create.
Dance	Entertainment	Specialized in a wide range of Chinese and Western dances, including folk, trendy, and classic dance moves, while also mastering advanced actions such as roundhouse kicks.
Lion Dance	Festival Performance	Focused on traditional folk performance, skillfully performing authentic Southern-style lion dance and showcasing the charm of traditional Chinese culture.

Safety Protection. When the robot enters an uncontrollable state, it automatically switches to DAMP mode to prevent damage. In dangerous situations, the joystick, App, and back-panel buttons can all trigger a soft emergency stop.

2. Specifications

2.1 Product Specifications

Type	Specification Parameters	Master Mini	Master Mini Pro	Master Mini Ultra
Basic Parameters	Height (when standing upright)	0.95m	0.95m	0.95m
	Weight	About 19.5kg	About 19.5kg	About 19.5kg
DoFs	Total DoFs	22	22	22
	Single Leg DoFs	6	6	6
	Single Arm DoFs	4	4	4
	Head DoFs	2	2	2
Operational Parameters	Walking Speed	1.1m/s	1.1m/s	1.1m/s
	Turning Speed	1.5rad/s	1.5rad/s	1.5rad/s
Battery Parameters	Battery Capacity	2Ah	5Ah	5Ah
	Battery Life	20min (1.1m/s)	1h10min (1.1m/s)	1h10min (1.1m/s)
	Charging Time	≤2h	≤2h	≤2h
	Cycle Life	≥500 times	≥500 times	≥500 times
Computing Platform	Computing	48 TOPS (Dense) AI performance	117 TOPS AI performance	200 TOPS AI performance
Sensors	Camera	Depth Camera	Depth Camera	Depth Camera
	Microphone	Microphone Array	Microphone Array	Microphone Array

	Speaker	1	1	1
Interaction interface	Buttons	Power×1, Interaction*3	Power×1, Interaction*3	Power×1, Interaction*3
	Indicator Light	RGB LED*1	RGB LED*1	RGB LED*1
Safety Fuction	Auditory Alerts	Low Battery Alert, Joint Overheat Alert	Low Battery Alert, Joint Overheat Alert	Low Battery Alert, Joint Overheat Alert
Communication Methods	Wired connection	Gigabit Ethernet	Gigabit Ethernet	Gigabit Ethernet
	Wireless Connection	WIFI 6	WIFI 6	WIFI 6
	Bluetooth	Bluetooth 5.2	Bluetooth 5.2	Bluetooth 5.2
	Certification Standards	CE/FCC	CE/FCC	CE/FCC
Noise	Walking Noise	≤70dB	≤70dB	≤70dB
Environmental Adaptability	Environmental Adaptability	-10℃~45℃	-10℃~45℃	-10℃~45℃
	Operating Humidity	5%~90%, without condensation	5%~90%, without condensation	5%~90%, without condensation

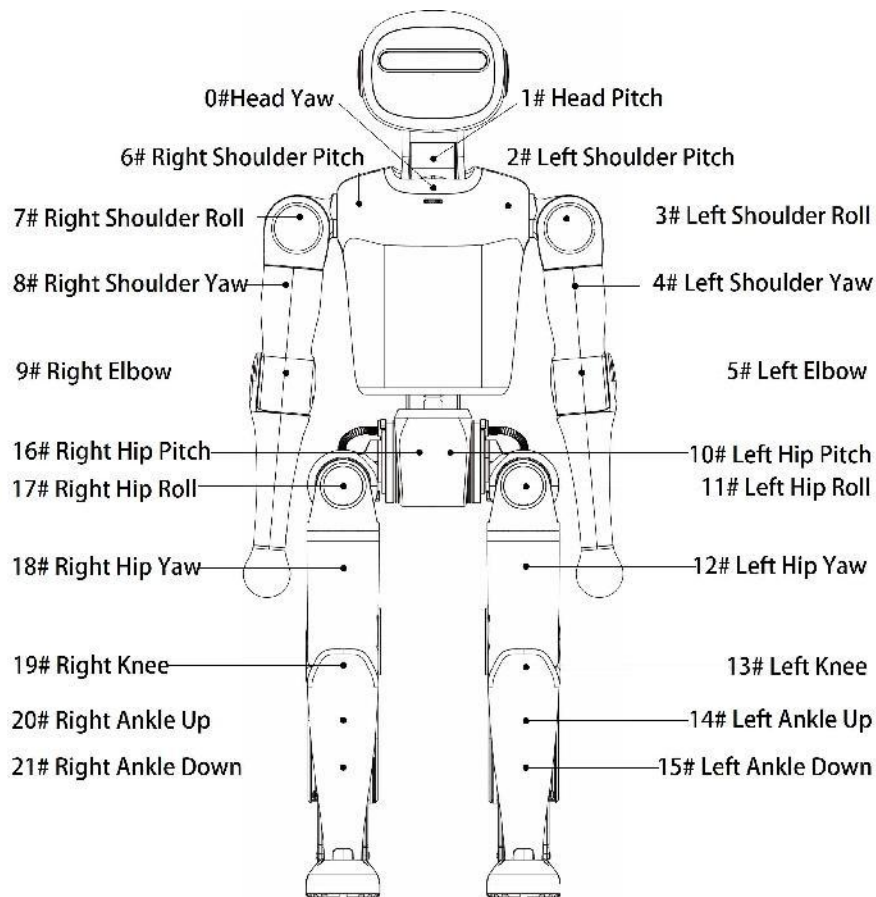
2. 2 Main Components

2.2.1 Joint Motors

2.2.1.1Joint IDs and Limits

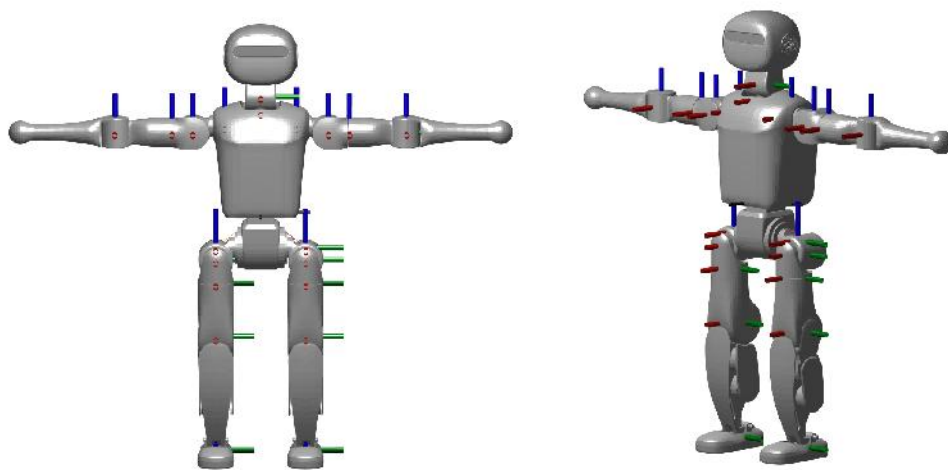
Joint ID	Joint Name	Limit (Degrees)	
		Max	Min
0	Head Yaw Joint	59	-59
1	Head Pitch Joint	49	-19
2	Left Shoulder Pitch Joint	69	-169
3	Left Shoulder Roll Joint	94	-94
4	Left Shoulder Yaw Joint	109	-109
5	Left Elbow Joint	39	-129

6	Right Shoulder Pitch Joint	69	-169
7	Right Shoulder Roll Joint	94	-94
8	Right Shoulder Yaw Joint	109	-109
9	Right Elbow Joint	129	-39
10	Left Hip Pitch Joint	128	-170
11	Left Hip Roll Joint	89	-22
12	Left Hip Yaw Joint	59	-59
13	Left Knee Joint	133	0
14	Left Ankle Up Joint	38	-17
15	Left Ankle Down Joint	41	-16
16	Right Hip Pitch Joint	128	-170
17	Right Hip Roll Joint	22	-89
18	Right Hip Yaw Joint	59	-59
19	Right Knee Joint	133	0
20	Right Ankle Up Joint	38	-17
21	Right Ankle Down Joint	41	-16



2.2.1.2 Coordinate Systems

When all joints are at zero degrees, the coordinate systems of the joints are shown below.



2.2.2 Controller

	Master Mini	Master Mini Pro	Master Mini Ultra
Processor	High-performance ARM Processor	Jetson Orin NX 8GB	Jetson AGX Orin 32GB
Computing performance	1×GoldP@3.2GHz 4×Gold@2.8GHz 3×Silver@2.0GHz 48 TOPS (Dense)	6-core Cortex-A78AE CPU@2GHz AI performance: 117 TOPS	8-core Cortex-A78AE CPU@2.2GHz AI performance: 200 TOPS
Memory	8GB	8GB	32GB
Storage	128GB	512GB	512GB
Wired Network	1000M*1	1000M*1	1000M*1
Wireless Network	Wi-Fi 6*1	WIFI6*1	Wi-Fi 6*1
Audio	Microphone, speaker	Microphone, speaker	Microphone, speaker

3. Safety and Warnings

Whenever the robot moves, stands up, or performs actions, there is a risk of falling, collision, or personal injury. The following items are the most common real-world risk points. Read them before operating the robot for the first time.

Read Before Operation

1. Before entering WALK mode, always enter PREP mode first and make sure the robot is standing securely on the ground.
2. Do not lift the robot while it is moving.
3. During motion, make sure there are no people or obstacles within the robot's operating range to avoid injury or collision.
4. During motion, do not touch any part of the robot with your body except the handle.

3.1 Venue and Actions

Large full-body actions such as full-body dances should be performed on hard surfaces such as rubber, cement, or tile. Carpet has high resistance and can increase the risk of falling. Keep people away from the robot throughout the action. For detailed venue and environment requirements, see "Overview".

3.2 During Upgrade

When installing new software, the robot's motion-control program is stopped and the joints no longer output torque. Before upgrading, make sure the robot is well supported, for example lying flat on the ground, and then switch it to DAMP mode to prevent it from falling during the upgrade.

3.3 Additional Notes for Secondary Development

After entering CUSTOM mode, all joints, including the legs, are controlled by your code, and the robot can easily lose balance and fall. Use a hoist to protect the robot throughout the process, and validate algorithms in Webots / Isaac simulation before running them on the physical robot. Be especially careful with changes involving biped control. CUSTOM mode can only be entered from PREP mode or DAMP mode, and can only switch back to those two modes. For the full mode description, see "[Modes](#)".

3.4 FCC Compliance Statement

Product Name: **Master Mini**

Model: Master Mini

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions:

1. this device may not cause harmful interference, and
2. this device must accept any interference received, including interference that may cause undesired operation.

Responsible Party: AlphaBeta Tech LLC

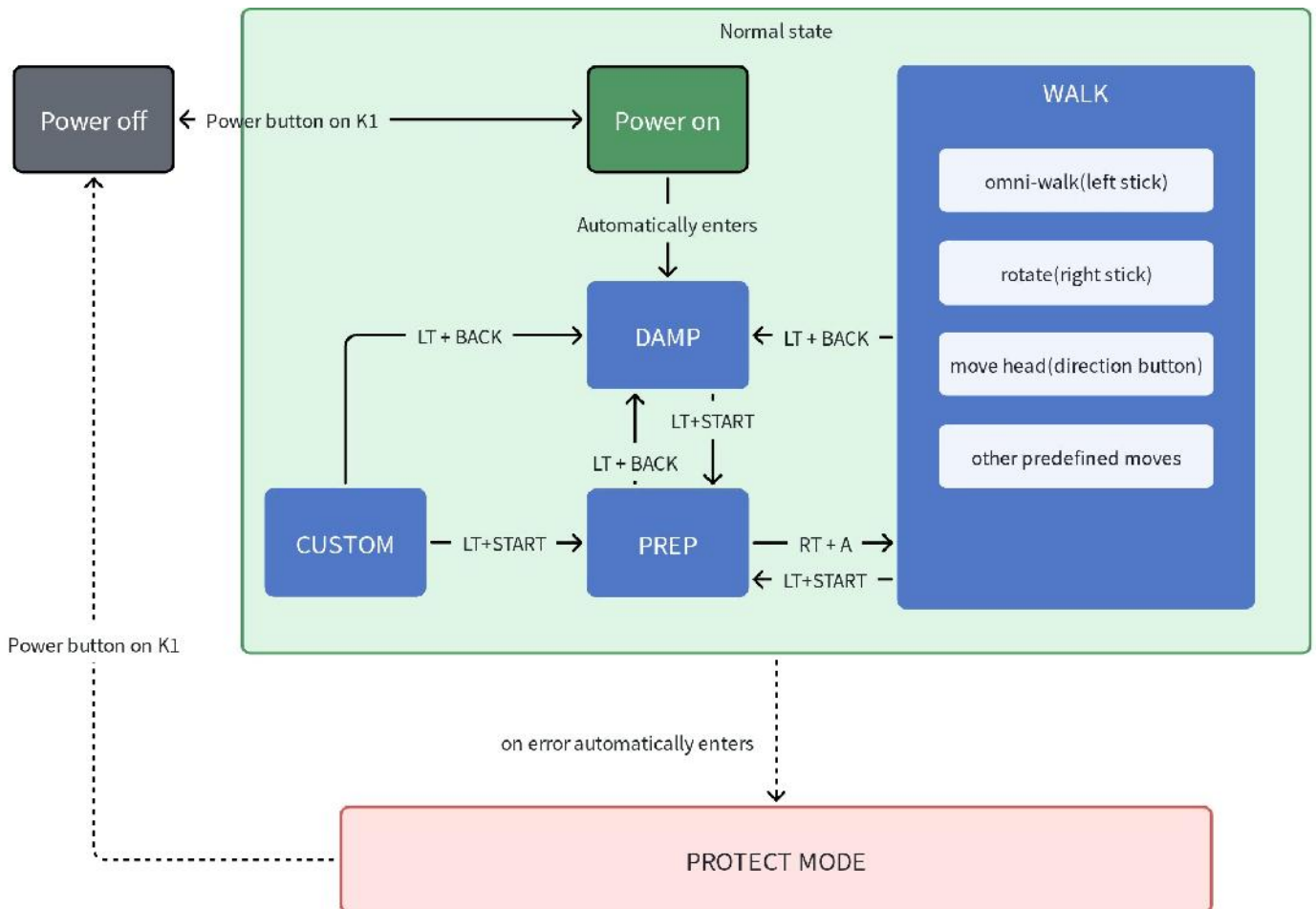
Address: 3080 OLCOTT ST UNIT B108 SANTA CLARA, CA 95054

Contact: alphabetallc@outlook.com

4. Modes

- The robot has multiple **mode**
- Different **mode** determines which **action** are available
- Different **mode** can switch between one another, but with certain constraints. For example, **DAMP** mode cannot switch directly to **WALK** mode; it must first enter **PREP** mode

4.1 State map



4.2 DAMP Mode (DAMP)

- The robot is powered on and the main controller is running normally.
- All joints enter a damped state. That is, when their positions are changed, there is **resistance** resistance; the joints do not actively change position and do not try to hold position.
- In DAMP mode, the robot cannot stand and needs effective support.
- DAMP mode is a safe mode and is less likely to cause accidental injury.
- DAMP mode can switch to **PREP mode** , but cannot switch directly to **WALK mode**

4.3 PREP Mode (PREP)

- The robot is powered on and the main controller is running normally.
- The robot enters and maintains a standing posture. That is, the robot assumes and holds a standing pose. If you try to change a joint position at this time, there is **very strong resistance** , and after displacement the joint will **actively recover** to return to the standing posture.
- In PREP mode, the robot can stand steadily and remain balanced on the ground. However, it cannot withstand large disturbances in this state because it only holds its posture and does not try to recover balance.
- PREP mode can switch to any other mode, including **DAMP Mode**、**WALK mode**

4.4 WALK Mode

- The robot is powered on and the main controller is running normally.
- The joints can execute corresponding preset **action** , including omnidirectional walking, turning, stepping, standing still, and head turning.
- Compared with **PREP mode**, **WALK mode** has stronger disturbance resistance and actively tries to recover balance after external forces within a certain range.
- WALK mode can switch to any other mode, including **DAMP mode** and **PREP mode**
- **Before entering this mode, make sure the robot is already in PREP mode and standing on flat ground.**

4.5 CUSTOM Mode

- The robot is powered on and the main controller is running normally.
- At this time, all robot joint control is **fully handed over to user-defined joint commands, including both legs. Use caution in this mode to prevent the robot from losing balance and falling.**
- CUSTOM mode can only be entered from PREP mode or DAMP mode.

- CUSTOM mode can only switch to DAMP mode or PREP mode.

⚠⚠ During secondary development, especially in CUSTOM mode, it is recommended to use a hoist to protect the robot throughout the process.

4.6 PROTECT Mode

- The robot automatically enters PROTECT mode when runtime issues occur, such as exceeding limits or falling.
- In PROTECT mode, the joint state is the same as **DAMP Mode** the same.
- PROTECT mode is a safe mode and is less likely to cause accidental injury.

5. Body Operations

There are four buttons on the MASTER MINI back panel:

1. Power button

- a. Press and hold for 3 seconds to power on; press and hold again for 3 seconds to power off.
- b. After powering off, wait 6 seconds before powering on again.

2. WLAK: Tap to enter WALK mode.

⚠⚠ Make sure the robot has switched to PREP mode and is standing steadily on the ground before switching to WALK mode.

3. **STAND:** Tap to enter PREP mode; the robot will fully straighten.
4. **F1:** customized button



6. Joystick Control

💡💡 Note that joystick button mappings vary by software version. Please confirm the current robot software version in the "Check Version" section.

6.1 Master Mini Joystick

6.1.1 Power on & off

- **Power On:** Press and hold the Home button for 5 seconds to power on. LED1, LED2, LED3 and LED4 all light up instantly upon startup. LED1 stays on steadily after boot completion.

- **Power Off:** Press and hold the Home button for 5 seconds to power off. LED1, LED2, LED3 and LED4 flash synchronously during shutdown. All indicator lights turn off after shutdown.

6.1.2 Charging

- Low battery: LED2 flashes as an alert.
- Charging in progress: LED2 keeps flashing.
- Fully charged: LED2 changes to steady on.

6.1.3 Basic Capabilities

Actions/Functions	Buttons	Prerequisites
Enter Master Mini Agent	L2 + R2 + A	
Enter Soccer Agent	L2 + R2 + B	
Enter Hi Chat Agent	L2 + R2 + X	This feature requires an internet connection
Enter Dance Agent	L2 + R2 + Y	
Enter Lion Dance Agent	L2 + R2 + UP	
Enter DAMP mode	L2 + Back	
Enter PREP mode	L2 + Start	
Enter WALK mode	R2 + A	Under PREP mode

locomotion control	Left stick	Under WALK mode
rotation control	Right stick	Under WALK mode
head rotation	Direction buttons	Under WALK mode

6.1.4 Agent Capabilities

Agent	Actions/Functions	Buttons	Prerequisites
Master Mini	Shake Hand	A	Under WALK mode
	Wave	B	Under WALK mode
	Ultraman	X	Under WALK mode
	Respect	Y	Under WALK mode
	Lucky Cat	R + A	Under WALK mode
	Dabbing	R + B	Under WALK mode
	Cheer	R + X	Under WALK mode
	Carry	R + Y	Under WALK mode
	CNY Dance	L + UP	Under WALK mode
	Rock Dance	L + LEFT	Under WALK mode
	Future Dance	L + RIGHT	Under WALK mode

	Get Up	L2 + UP	Under DAMP/PREP mode
Soccer	Shake Hand	A	Under WALK mode
	Wave	B	Under WALK mode
	Get Up	L2 + UP	Under DAMP/PREP mode
	Track Ball	L2 + A	Under WALK mode
	Chase Ball	L2 + B	Under WALK mode
	Kick Ball	L2 + X	Under WALK mode
	Power Kick	L2 + Y	Under WALK mode
	Siuuu	L + Y	Under WALK mode
	Arms Crossed	L + B	Under WALK mode
	Point to Sky	L + UP	Under WALK mode
	Cradle Dance	L + DOWN	Under WALK mode
Dance	Rock Dance	L+A	Under WALK mode
	CNY Dance	L+B	Under WALK mode
	Future Dance	L+X	Under WALK mode

	MJ Dance	L+Y	Under WALK mode
	Arabic Dance	L+DOWN	Under WALK mode
	Boxing Style Kick	L+LEFT	Under WALK mode
	Roundhouse Kick	L+RIGHT	Under WALK mode
	Mounts & Pals	L+R+A	Under WALK mode
	Reform Breeze	L+R+B	Under WALK mode
Hi Chat	Default	L2+A	Under WALK mode
	XiaoBao	L2+B	Under WALK mode
	Gentle Ella	L2+X	Under WALK mode
	Grumpy Jax	L2+Y	Under WALK mode
	English Tutor	L2+UP	Under WALK mode
	Cute Pet	L2+DOWN	Under WALK mode
	Alien Weirdo	L2+LEFT	Under WALK mode
	Say Hello	R2 + Y	Enable chat
Lion Dance	Dance Prepare	L+A	Under DAMP/PREP mode

	Dance Walk	L+B	Under WALK mode
	Lion Dance 1	L+X	Under WALK mode
	Lion Dance 2	L+Y	Under WALK mode

6.1.5 Xbox Joystick

Use a standard Xbox joystick. **Before use, confirm that the joystick control mode is receiver mode and that the operating mode shows three lights steadily on.**

6.1.6 Basic Capabilities

Actions/Functions	Buttons	Prerequisites
Enter Master Mini Agent	LT + RT + A	
Enter Soccer Agent	LT + RT + B	
Enter Hi Chat Agent	LT + RT + X	
Enter Dance Agent	LT + RT + Y	
Enter Lion Dance Agent	LT + RT + UP	
Enter DAMP mode	LT + Back	
Enter PREP mode	LT + Start	
Enter WALK mode	RT + A	Under PREP mode

locomotion control	Left stick	Under WALK mode
rotation control	Right stick	Under WALK mode
head rotation	Direction buttons	Under WALK mode

6.1.7 Agent Capabilities

Agent	Actions/Functions	Buttons	Prerequisites
Master Mini	Shake Hand	A	Under WALK mode
	Wave	B	Under WALK mode
	Ultraman	X	Under WALK mode
	Respect	Y	Under WALK mode
	Lucky Cat	RB + A	Under WALK mode
	Dabbing	RB + B	Under WALK mode
	Cheer	RB + X	Under WALK mode
	Carry	RB + Y	Under WALK mode
	CNY Dance	LB + UP	Under WALK mode
	Rock Dance	LB + LEFT	Under WALK mode
	Future Dance	LB + RIGHT	Under WALK mode

	Get Up	LT + UP	Under DAMP/PREP mode
Soccer	Shake Hand	A	Under WALK mode
	Wave	B	Under WALK mode
	Get Up	LT + UP	Under DAMP/PREP mode
	Track Ball	LT + A	Under WALK mode
	Chase Ball	LT + B	Under WALK mode
	Kick Ball	LT + X	Under WALK mode
	Power Kick	LT + Y	Under WALK mode
	Siuuu	LB + Y	Under WALK mode
	Arms Crossed	LB + B	Under WALK mode
	Point to Sky	LB + UP	Under WALK mode
	Cradle Dance	LB + DOWN	Under WALK mode
Dance	Rock Dance	LB+A	Under WALK mode
	CNY Dance	LB+B	Under WALK mode
	Future Dance	LB+X	Under WALK mode

	MJ Dance	LB+Y	Under WALK mode
	Arabic Dance	LB+DOWN	Under WALK mode
	Boxing Style Kick	LB+LEFT	Under WALK mode
	Roundhouse Kick	LB+RIGHT	Under WALK mode
	Mounts & Pals	LB+RB+A	Under WALK mode
	Reform Breeze	LB+RB+B	Under WALK mode
Hi Chat	Default	LT+A	Under WALK mode
	XiaoBao	LT+B	Under WALK mode
	Gentle Ella	LT+X	Under WALK mode
	Grumpy Jax	LT+Y	Under WALK mode
	English Tutor	LT+UP	Under WALK mode
	Cute Pet	LT+DOWN	Under WALK mode
	Alien Weirdo	LT+LEFT	Under WALK mode
	Say Hello	RT + Y	Enable chat
Lion Dance	Dance Prepare	LB+A	Under DAMP/PREP mode

	Dance Walk	LB+B	Under WALK mode
	Lion Dance 1	LB+X	Under WALK mode
	Lion Dance 2	LB+Y	Under WALK mode
	Lion Dance 3	LB+UP	Under WALK mode

7. Connect to Robot

1. Use Agents

8.1 What Is an Agent?

An Agent is a "skill pack" installed in the robot's brain. It can also be understood as an application running on the robot. Like switching apps on a phone, switching Agents lets the robot enter the corresponding capability scenario and use different interaction methods and task capabilities.

8.2 How to Switch Agents

1. Switch via the mobile App

- a. Tap the Agent name in the top-left corner to switch to another Agent. Each Agent has different capabilities — for example, Soccer Master can play soccer with you, Hi Chat can talk with you using different personas, and Dance Master can perform a variety of dances from around the world.

2. Switch via the Joystick

Switch Agents	Master Mini joystick buttons	Xbox joystick buttons
Enter Master Mini Agent	L2 + R2 + A	LT + RT + A
Enter Soccer Agent	L2 + R2 + B	LT + RT + B
Enter Hi Chat Agent	L2 + R2 + X	LT + RT + X
Enter Dance Agent	L2 + R2 + Y	LT + RT + Y
Enter Lion Dance Agent	L2 + R2 + UP	LT + RT + UP

8.3 Official Agents

Master Mini includes 6 official Agents covering entertainment, conversation, teaching, competition, and other scenarios.

8.3.1 Master Mini

A general-purpose basic Agent that supports daily actions such as waving, handshaking, and bowing with folded hands. It can also perform several fun dances for everyday interaction scenarios.

Video1, local

Functions list

Functions	Master Mini joystick buttons	Xbox joystick buttons	Prerequisites
Shake Hand	A	A	Under WALK mode
Wave	B	B	Under WALK mode
Ultraman	X	X	Under WALK mode
Respect	Y	Y	Under WALK mode
Lucky Cat	R + A	RB + A	Under WALK mode
Dabbing	R + B	RB + B	Under WALK mode
Cheer	R + X	RB + X	Under WALK mode
Carry	R + Y	RB + Y	Under WALK mode
CNY Dance	L + UP	LB + UP	Under WALK mode

Rock Dance	L + LEFT	LB + LEFT	Under WALK mode
Future Dance	L + RIGHT	LB + RIGHT	Under WALK mode
Get Up	L2 + UP	LT + UP	Under DAMP/PREP mode

8.3.2 Soccer

Designed for soccer scenarios. It can autonomously track, chase, kick, and shoot the ball, and can accurately reproduce signature celebration moves from soccer stars such as Cristiano Ronaldo and Messi.

Video2, local

Functions list

Functions	Master Mini joystick buttons	Xbox joystick buttons	Prerequisites
Shake Hand	A	A	Under WALK mode
Wave	B	B	Under WALK mode
Get Up	L2 + UP	LT + UP	Under DAMP/PREP mode
Track Ball	L2 + A	LT + A	Under WALK mode
Chase Ball	L2 + B	LT + B	Under WALK mode
Kick Ball	L2 + X	LT + X	Under WALK mode
Power Kick	L2 + Y	LT + Y	Under WALK mode
Siuuu	L + Y	LB + Y	Under WALK mode
Arms Crossed	L + B	LB + B	Under WALK mode
Point to Sky	L + UP	LB + UP	Under WALK mode
Cradle Dance	L + DOWN	LB + DOWN	Under WALK mode

8.3.3 Hi Chat

An intelligent conversational Agent that can switch flexibly among fun personas such as a cute child, tutor, or pet, meeting needs for casual chat and entertaining companionship.

See details in [Voice Interaction](#)。

Video 3, local

Functions list

Functions	Master Mini joystick buttons	Xbox joystick buttons	Prerequisites
Default	L2 + A	LT + A	Under WALK mode
XiaoBao	L2 + B	LT + B	Under WALK mode
Gentle Ella	L2 + X	LT + X	Under WALK mode
Grumpy Jax	L2 + Y	LT + Y	Under WALK mode
English Tutor	L2 + UP	LT + UP	Under WALK mode
Cute Pet	L2 + DOWN	LT + DOWN	Under WALK mode
Alien Weirdo	L2 + LEFT	LT + LEFT	Under WALK mode
Say Hello	R2 + Y	RT + Y	Enable chat

8.3.4 Teach

A hands-on learning Agent. Users physically guide the robot through a motion, and the device quickly recognizes and accurately reproduces it, making custom actions easy to create.

Video 4, local

8.3.5 Dance

Specialized in a wide range of Chinese and Western dances, including folk, trendy, and classic dance moves, while also mastering advanced actions such as roundhouse kicks.

❗❗ Full-body dances require a dedicated venue. Rubber, cement, or tile floors are recommended; high-resistance surfaces such as carpet increase the risk of falling.

Video 5, local

Functions list

Functions	Master Mini joystick buttons	Xbox joystick buttons	Prerequisites
Rock Dance	L + A	LB + A	Under WALK mode
CNY Dance	L + B	LB + B	Under WALK mode
Future Dance	L + X	LB + X	Under WALK mode
MJ Dance	L + Y	LB + Y	Under WALK mode
Arabic Dance	L + DOWN	LB + DOWN	Under WALK mode
Boxing Style Kick	L + LEFT	LB + LEFT	Under WALK mode
Roundhouse Kick	L + RIGHT	LB + RIGHT	Under WALK mode
Mounts & Pals	L + R + A	LB + RB + A	Under WALK mode
Reform Breeze	L + R + B	LB + RB + B	Under WALK mode

8.3.6 Lion Dance

Focused on traditional folk performance, skillfully performing authentic Southern-style lion dance and showcasing the charm of traditional Chinese culture.

Functions list

Functions	Master Mini joystick buttons	Xbox joystick buttons	Prerequisites
Dance Prepare	L + A	LB + A	Under DAMP/PREP mode
Dance Walk	L + B	LB + B	Under WALK mode
Lion Dance 1	L + X	LB + X	Under WALK mode

Lion Dance 2	L + Y	LB + Y	Under WALK mode
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2. Service Start/Stop

9.1 Use the App

1. On the control page, tap "Settings" to enter the settings page.
2. On the settings page, tap "Restart Robot" to restart the robot.

9.2 Use a Terminal

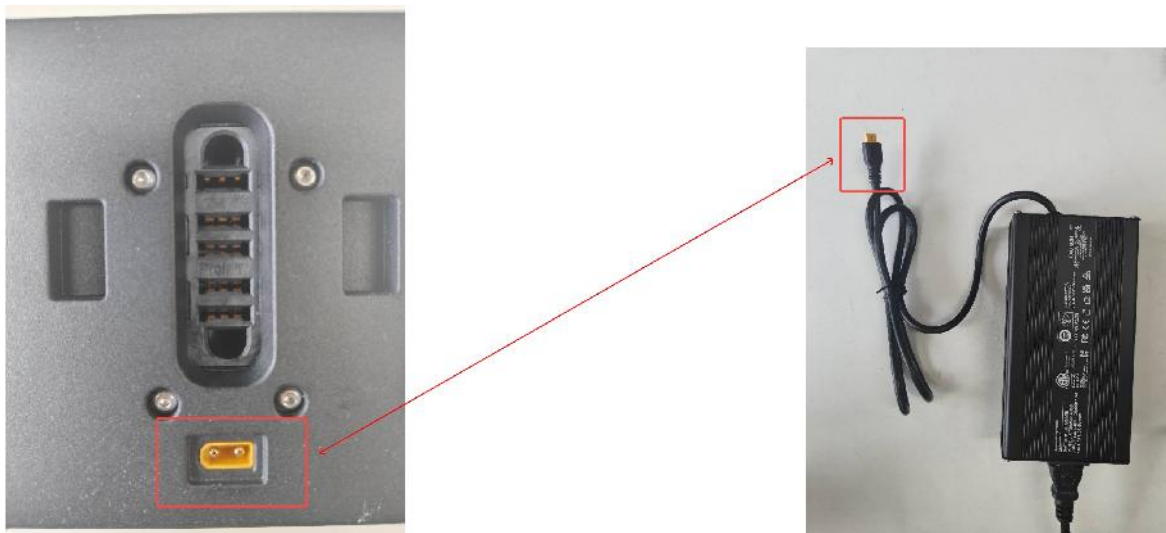
1. Use another device to [connect to the robot through a terminal](#)
2. Run the following commands in any directory on the robot:

💡💡 Note: restarting the robot service fully restarts the robot firmware. Make sure the robot is properly supported on a stand or lying flat on the ground before doing this.

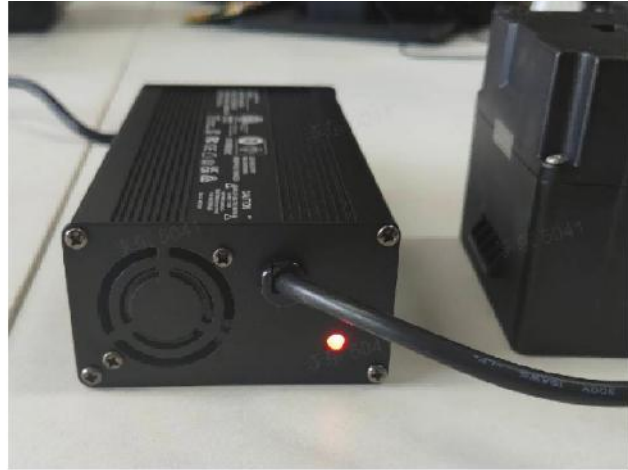
3. Charging

10.1 5Ah Battery Charging

1. Insert the charging plug according to the indicated direction.

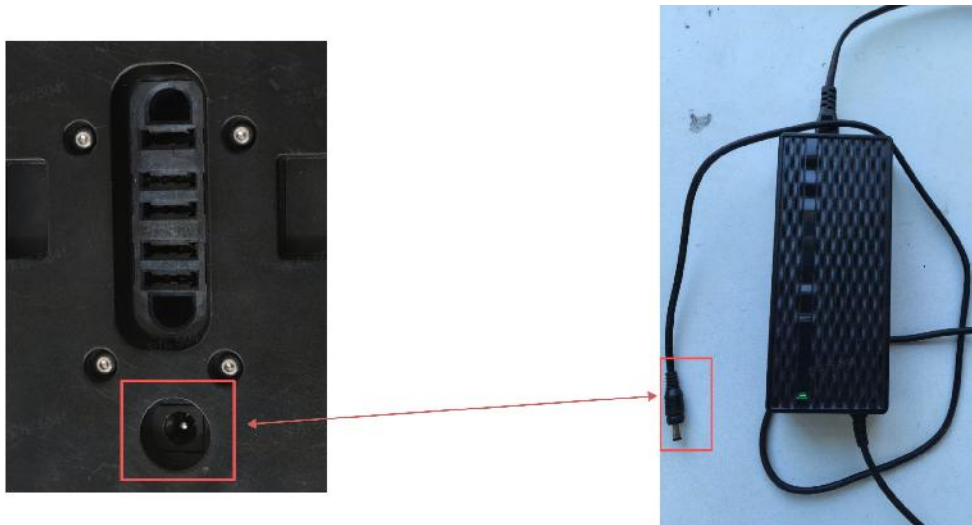


2. The charging status is shown as follows: the green light indicates a full charge, and the red light indicates charging in progress.



10.2 2Ah Battery Charging

1. Insert the charging plug according to the indicated direction.



2. The charging status is shown as follows: the green light indicates a full charge, and the red light indicates charging in progress.



4. Voice Interaction

12.1 Introduction

12.1.1 Hi Chat Agent

For v1.3.1.1 and above, the **Master Mini** robot is equipped with AI real-time voice dialogue.

This feature can be enabled or disabled via the “Hi Chat” Agent in the mobile app.

12.1.1.1 Usage Instructions

- Open the “Hi Chat” entry in the mobile app, tap and select a persona. When you hear a voice prompt similar to “Hello, I am the **Master Mini** robot. How can I assist

you?” , the real-time voice dialogue function is activated, and you can start talking to the robot.

- Once real-time voice dialogue is enabled, the robot will automatically turn its head to face the nearest person in front of it.
- AI voice interaction requires an internet connection.

12.1.1.2 Voice Commands

Note: When using a custom persona, conflicting system prompts may cause voice commands to fail.

Master Mini robots currently support voice commands that make the robot perform corresponding actions. The large model may also autonomously trigger the following actions based on the current conversation context. The action set includes:

Action Category	Command Name	Trigger Condition
Basic Movement	Turn Left	Triggered when explicit commands such as "turn left" or "turn to the left" are received.
	Turn Right	Triggered when explicit commands such as "turn right" or "turn to the right" are received.
	Turn Left One Full Circle	Triggered when commands such as "turn left one full circle" or "turn counterclockwise one full circle" are received.
	Turn Right One Full Circle	Triggered when commands such as "turn right one full circle" or "turn clockwise one full circle" are received.
	Walk Forward	Triggered when commands such as "walk forward" or "go forward" are received.
	Walk Backward	Triggered when commands such as "walk backward" or "move back" are received.
	Move Left	Triggered when commands such as "move left" or

		"move to the left" are received.
	Move Right	Triggered when commands such as "move right" or "move to the right" are received.
Interaction Actions	Wave	Triggered when commands such as "wave" or "wave your hand" are received, or when expressing goodbye.
	Handshake	Triggered when explicit social commands such as "shake hands" or "let's shake hands" are received.
	Greet	Triggered only when the user explicitly greets the robot, such as "hello" or "Hi MASTER MINI"; the reply should include a greeting.
	Nod	Triggered when the robot needs to express agreement or approval, such as responding to "the weather is nice today" or "did I do it right?"; the reply may use an affirmative tone.
	Shake Head	Triggered when the user asks about unknown content, such as "what does this word mean?"; execute this action if the knowledge base has no answer.

13. Firmware Version

13.1 Check Version

13.1.1 Check via App

After connecting to the robot, tap "Settings - About" to view the robot's current version.

13.1.2 Check via Terminal

After connecting to the robot's board, run the following command to view the current system version of the robot.

13.2 Version Upgrade

13.2.1 Before Upgrading

When installing a new version of software, the robot's motion control program will be stopped, and the robot's joints will not exert force. **Before upgrading, ensure the robot is in damping mode or motion control is stopped and has good support (e.g., placing it flat on the ground or using a stand to support the robot). Do not cut off power or interrupt the process during the upgrade.**

- Make sure the installation package matches the current MASTER MINI model and the target version.
- Make sure the robot's battery level, network, and terminal connection are stable.
- If you have modified any configuration files, back up important configurations and logs first.

13.2.2 Upgrade via App

In the robot settings, tap "Firmware Update".

- Before installing the firmware, make sure the robot has entered damping or ready mode, and the battery level is above 50%.

13.2.3 FAQ

13.2.3.1 Slow pip Downloads

The perception process depends on some Python libraries installed via pip. If downloads are slow, you can switch the pip download source and run the upgrade again.

```
sudo su
```

```
pip config set global.index-url https://pypi.mirrors.ustc.edu.cn/simple
```

```
./v1.1.1.0-release-aarch64.run
```

13.2.3.2 apt update Fails

If the upgrade log indicates that apt update failed, you can use the fishros tool to switch sources and try the upgrade again.

```
# Recommended: use fishros and select in order:
```

```
# [5] One-click source change
```

```
# [1] Change system source only
```

```
# [1] Add ROS/ROS2 source
```

```
wget http://fishros.com/install -O fishros && . fishros
```

14. Troubleshooting and Support

14.1 Restore Factory Settings

When MASTER MINI cannot return to a usable state due to configuration changes, software errors, or environment changes, you can restore factory settings to reinstall the factory software.

14.1.1 When to Use

- Abnormal behavior after modifying the robot's configuration, which cannot be recovered through normal configuration rollback.
- Technical support recommends reinstalling the factory software to rule out environment issues.
- You need to restore the robot to a software state close to its initial delivery.

14.1.2 Before You Begin



Restoring factory settings may erase existing configurations and local changes. Before proceeding, back up important configurations, logs, and custom files, and make sure the robot is in a stable, well-supported state.

- Confirm that the current robot model and the recovery package are compatible with MASTER MINI.
- Make sure the power, network, and terminal connection are stable, and do not cut off power during the recovery process.
- If you are not sure whether a recovery is needed, contact technical support first.

14.1.3 Perform the Recovery

After logging in to the robot's board, navigate to the recovery directory and run the recovery program. The following command uses the installation package name in the current recovery directory; if there are multiple packages in the directory, use the file that matches your device version.

14.1.4 After the Recovery

1. Wait for the recovery program to finish, and restart the robot as prompted.
2. Confirm that the robot starts normally and that basic services and network connection are restored.
3. Reconfigure the network, App, voice, or custom parameters that you want to keep.
4. If the robot still behaves abnormally after recovery, record the symptoms and provide logs to technical support as described on the "Log Retrieval" page.

14.2 Log Retrieval

When MASTER MINI has issues, follow the steps on this page to retrieve logs and send them to technical support. Logs help locate when the issue occurred, the running state, and related error information.

14.2.1 Privacy and Purpose

Our company will, without violating any applicable laws and regulations or involving any personal privacy information, collect only data related to the operation of the robots for the purposes of fault detection, problem localization, and statistical analysis.

14.2.2 Upload Logs via App

1. After connecting to the robot via the App, go to "Settings".
2. Tap "Log Upload".
3. Select the time range to upload, making sure it covers the time when the issue occurred.
4. The upload takes about 5-10 minutes. During this process, keep the robot powered on and connected to the network. Both the App logs and the robot logs will be uploaded together.
5. After the upload is complete, contact customer service to report the issue. Our technical staff will retrieve your uploaded log files through the management console.

14.2.3 Retrieve Logs via Terminal

Before you start, record the time the issue occurred, the symptoms, and the operation you were performing at the time as accurately as possible. The robot uses the UTC+8 time zone by default; if you are in a different time zone, convert the time before filling in the log time range.

1. [Log in to the robot via terminal.](#)

2. Confirm the time the issue occurred and prepare the start and end times for the logs.

The time format is YYYYMMDD-HHMMSS.

3. Run the log export command.
4. Select a range that covers the time before and after the issue. For example, if the issue occurred around 20200808-120810, you can export logs about 10 seconds before and after:
5. **Send the log** package to technical support, along with the symptoms, the time of occurrence, and the reproduction steps.

14.3 Remote Support

14.3.1 Introduction

When the robot has issues and you need remote assistance, follow these steps to get remote support from our staff.

14.3.2 Information to Send to Technical Support

- Device ID and one-time password.
- The robot's current status: whether it can power on, whether it can connect to the network, and its current mode.
- Symptoms: for example, unable to walk, abnormal App connection, upgrade failure, etc.
- The time the issue occurred and the operation you were performing at the time.
- Whether logs have been uploaded, or exported by yourself via terminal.

14.3.3 Remote Assistance Procedure



You can choose from the following two remote assistance methods:

1. Connect your personal computer to the robot via SSH; technical support can then connect to your computer via remote desktop and subsequently to the robot via SSH.
2. Install remote desktop software on the robot; you can use a display and keyboard/mouse to open the remote desktop software, allowing support personnel to connect directly to the robot.